

House Price Prediction Using Linear Regression

Introduction

House price prediction is one of the most common applications of Machine Learning in the real estate industry. Predicting the value of a house based on its characteristics helps buyers, sellers, and real estate agencies make informed decisions. In this project, a Linear Regression model is developed to predict house prices using selected housing features from the Kaggle House Prices dataset.

Problem Statement

The objective of this project is to build a Machine Learning model that can predict the selling price of a house based on its living area, number of bedrooms, and number of bathrooms. Linear Regression is used to establish the relationship between these housing features and the final sale price.

Dataset Description

The dataset used for this project is the House Prices: Advanced Regression Techniques dataset from Kaggle. It contains information about residential houses and their sale prices.

The following features were selected for this task:

- GrLivArea: Above-ground living area in square feet.
- BedroomAbvGr: Number of bedrooms above ground level.
- FullBath: Number of full bathrooms.
- SalePrice: Actual selling price of the house (target variable).

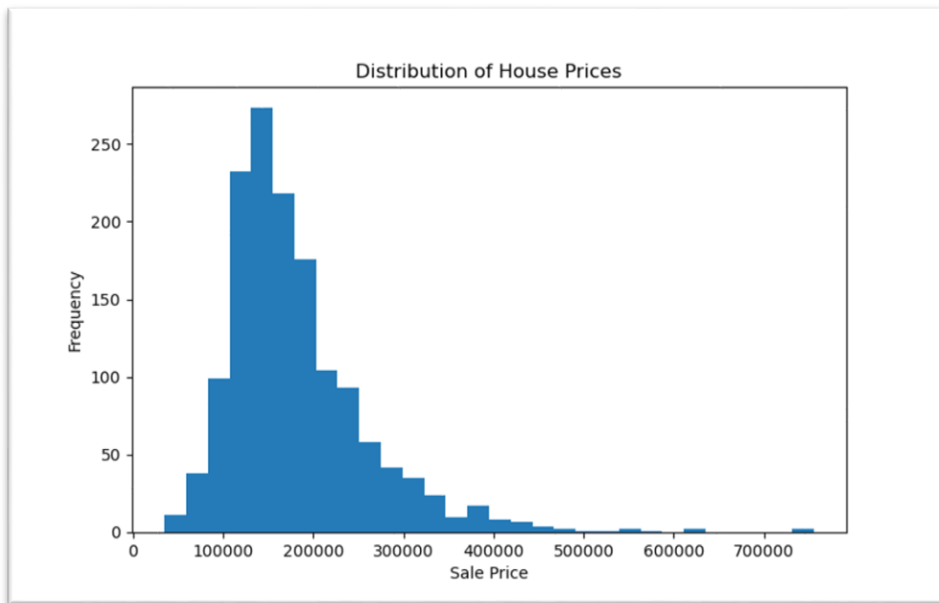
These features were chosen because they have a direct influence on the value of a house and are easy to interpret.

Data Exploration and Visualization

Before building the model, the dataset was explored to understand its structure and characteristics. Missing values were checked, and the selected features were analyzed.

Several visualizations were created:

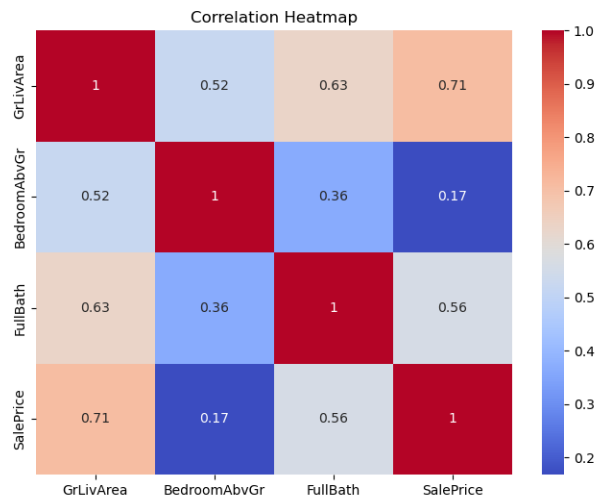
1. Distribution of House Prices – to understand how sale prices are distributed.



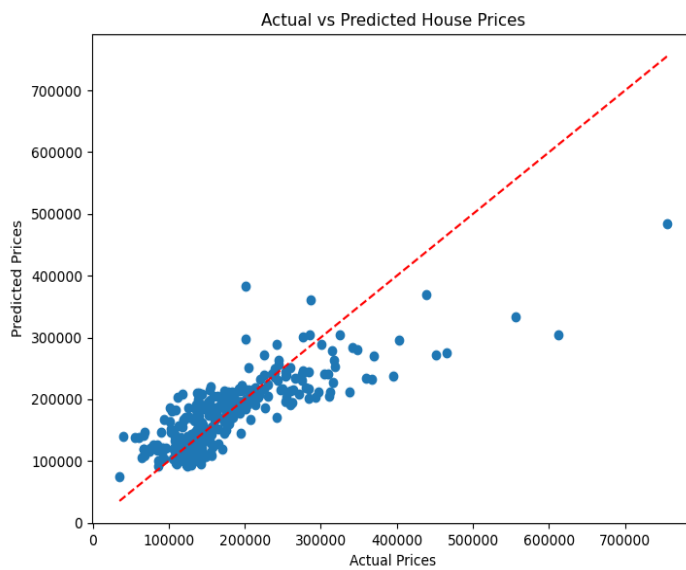
2. Living Area vs Sale Price – to observe the relationship between house size and price.



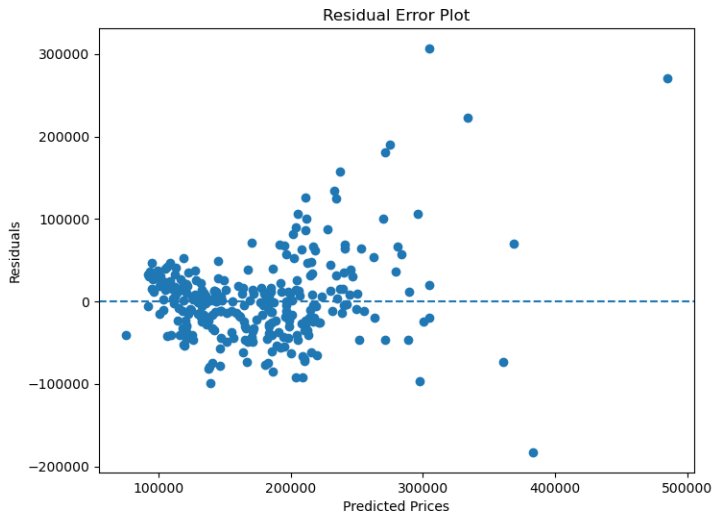
3. Correlation Heatmap – to identify relationships among features.



4. Actual vs Predicted Plot – to compare model predictions with actual values.



5. Residual Plot – to analyze prediction errors.



These visualizations provide valuable insights into the dataset and help evaluate model performance.

Methodology

The project followed the following steps:

1. Load the dataset using Pandas.
2. Select relevant features and target variable.
3. Perform exploratory data analysis and visualization.
4. Split the dataset into training and testing sets.
5. Train a Linear Regression model using the training data.
6. Generate predictions on the test data.
7. Evaluate model performance using standard regression metrics.

Algorithm Used

Linear Regression is a supervised machine learning algorithm used for predicting continuous numerical values. It establishes a linear relationship between independent variables (features) and a dependent variable (target).

The model attempts to find the best-fitting line that minimizes prediction errors. In this project, the algorithm learns how living area, bedrooms, and bathrooms influence house prices and uses that relationship to make future predictions.

Evaluation Metrics

The performance of the model was evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

These metrics help determine how accurately the model predicts house prices.

Results

The Linear Regression model was successfully trained and tested on the dataset. The evaluation metrics indicated that the model was able to capture the relationship between housing features and sale prices. Among the selected features, living area showed the strongest influence on house price prediction.

The generated visualizations further confirmed the effectiveness of the model and provided insights into feature relationships and prediction accuracy.

Conclusion

This project successfully implemented a Linear Regression model for house price prediction. By using housing characteristics such as living area, number of bedrooms, and number of bathrooms, the model was able to estimate house prices with reasonable accuracy. The project demonstrates the practical application of Machine Learning techniques in solving real-world prediction problems and provides a foundation for developing more advanced predictive models in the future.